MAYANK AGRAWAL

Full Stack, Backend & AI Systems Engineer

+91 9340861338 | agrawalmayank1111@gmail.com | linkedin.com/in/mayank-agrawal | github.com/magraa | magraa.github.io

PROFESSIONAL SUMMARY

Software Developer with extensive production experience building scalable web applications and RESTful APIs using the MERN stack and Next.js. Strong expertise in backend architectures, cloud integrations, and AI-driven automation systems. Experienced in shipping end-to-end commercial products, reducing latency, and building reliable 60fps user interfaces. IIIT Gwalior alumnus.

WORK EXPERIENCE

Software Developer · *Bittern Technologies (Remote)* — **Apr 2024 – Dec 2025**

- Architected and deployed scalable full-stack web applications using the MERN stack, Next.js, and cloud infrastructure.
- Engineered and optimized RESTful APIs and Node.js microservices, cutting database roundtrip latency by 35%.
- Built AI-driven automation pipelines and data extraction tools using JavaScript and Python, slashing operational overhead.
- Engineered offline-first state synchronization and reactive store layers, guaranteeing fluid 60fps UI transitions across viewports.

Software Engineer Intern · *Logical Analytics (Remote, Lucknow)* — **Sep 2023 – Dec 2023**

- Built responsive analytics dashboards and reusable UI component libraries using React.js and SCSS.
- Designed backend schemas and RESTful APIs using Node.js, Express, and MongoDB collections.
- Integrated Cloudinary media transformation pipelines, cutting server asset bandwidth and storage consumption by 30%.

SaaS Developer · *Blockchain Startup & Clients (Remote)* — **Nov 2022 – Feb 2023**

- Developed blockchain automation bots, alert systems, and real-time transaction event listeners using Python and Discord.py.
- Built automated web interaction bots using Selenium, Minimax algorithms, and custom licensing validation systems.

FEATURED PRODUCTION PROJECTS

Shoboti – Social Video Commerce Platform · *shoboti.com* | *aria.shoboti.com* — **Next.js 15, TypeScript, PostgreSQL, Drizzle ORM, Gemini AI**

- Built an end-to-end social e-commerce engine converting Instagram comments into instant personalized DMs with rotating anti-spam protection.
- Created conversational shopping assistant with pgvector semantic retrieval and powered live mobile storefronts with shoppable reels.

Hisab AI – Financial Notebook & Bahi-Khata PWA · *hisab-ai-livid.vercel.app* — **Next.js 15, React 19, Supabase RLS, Gemini 2.0 Flash, PWA**

- Engineered offline-capable PWA ledger parsing crumpled receipts with Gemini 2.0 Flash multimodal OCR with 95%+ precision.
- Implemented 1-tap client-side vector PDF generation with Rupee formatting and direct WhatsApp dispatch.

SIMVox-AI – Zero-VoIP Cellular AI Voice Agent · *github.com/Magraa/SIMVox-AI* — **Python, Gemini LLM, Edge Neural TTS, ADB, PyWebView (10★ Stars)**

- Bridges PC to Android cellular hardware over ADB, converting handsets into autonomous receptionists with zero VoIP/SIP fees.
- Pipes audio to Google Gemini and synthesizes answers via Edge Neural TTS with sub-second turnaround latency.

Mr. Proper Platform & Proper Perks · *mrproper.in* — **Next.js 14, Supabase, Tailwind, Razorpay, Google Maps API**

- Shipped full-stack e-commerce, contractor loyalty portal, GPS-logged field CRM, and formula batching ERP (95+ Lighthouse).
- Engineered automated Google Maps distance-matrix technician routing and Razorpay webhook reconciliation, eliminating manual dispatch latency.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, C/C++
Web & Backend: React.js, Next.js (App Router), Node.js, Express.js, RESTful APIs, MongoDB, Supabase (RLS), WebSockets, Tailwind CSS v4
Cloud, DevOps & AI: AWS (EC2/S3), Docker, Git/GitHub, Vercel, Google Gemini API, LangChain, Multimodal OCR Vision, Edge-TTS, Postman

EDUCATION

IIIT Gwalior
Integrated B.Tech + MBA (IT) · 2019 – 2024
Institute of National Importance

ACHIEVEMENTS & CERTS

- Solved 300+ DSA problems (LeetCode, HackerRank).
- Pre-finalist at Hacksagon 2020 national hackathon.
- Complete 2023 Web Dev Bootcamp (Dr. Angela Yu).